MIL-C-26165D(USAF)
26 May 1972

SUPERSEDING
MIL-C-26165C(USAF)
4 May 1962

MILITARY SPECIFICATION

COMPRESSOR, AIR, ELECTRIC MOTOR DRIVEN, 30 CFM, 400 PSI, SKID MOUNTED TYPE AF/E32A-1

1. SCOPE

1.1 Scope. This specification covers a 30 cfm, 400 psi, electric motor driven, skid mounted, explosion-proof air compressor designated AF/E32A-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-3600	Compressors, Air and Gas (Except Oxygen), Packaging of
MIL-H-8795	Hose Assembly, Hydraulic, Fuel and Oil Resistant
MIL-C-26805	Compressor Units, Air/Gas, General Requirements for

STANDARDS

Federal

Fed. Std. No. 595	Colors

Military

MIL-STD-130	Identification Marking for U. S. Military Property

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3.4.3 Gages. Gages shall be provided to indicate the pressure in the receiver and at the outlet of the pressure regulator. The gages shall cover the range of 0 to 600 psi with increments of not more than 10 psi and shall be red-lined at 500 psi. The gages shall have a dial diameter of not less than 4½ inches and shall be accurate to within ± 2 percent of full-scale reading.

3.4.4 Compressor pressure regulation.

3.4.4.1 Constant-speed unloader. The range between loading and unloading shall not exceed 75 psi. The unloader shall be adjustable over a range of 150 to 500 psi. The compressor shall be set to unload at 425 ± 10 psi.

3.4.4.2 Start-stop control. The start-stop control shall be set to stop the compressor when the pressure reaches 400 ± 10 psi and to start the compressor when the pressure drops to 340 ± 10 psi.

3.4.5 Moisture separators. Moisture and oil separators shall be provided after each intercooler and the aftercooler to remove all condensed moisture and oil. At an ambient temperature of 100° F and a relative humidity of 100 percent, the separators shall be either large enough to permit continuous operation of the compressor unit for a minimum period of 90 minutes without unloading, or capable of automatically draining the separators when required. A separate intercooler and aftercooler moisture separator drain valve shall be provided on the instrument panel to permit manual draining of the intercooler and aftercooler moisture separators.

3.4.6 Receiver. A receiver shall be provided. It shall be designed for an operating pressure of 500 psi and shall have a minimum volume of 16 cubic feet. The receiver shall be so located that all moisture, oil or other foreign matter will drain toward the receiver drain valve connection. The receiver shall be removable for servicing and testing and it shall be possible to inspect the receiver for fault and corrosion without removal.

3.4.7 Servicing system. The servicing system plumbing shall be such that by closing one valve the receiver can be bypassed.

3.4.7.1 Servicing hose. A 30-foot long size 4 hose conforming to MIL-H-8795 shall be provided. The hose shall be mounted on a live hose reel which shall be spring motor operated to automatically rewind the hose when released from any lock position up to and including the specified length of hose.

3.4.7.2 Servicing valve. A servicing valve shall be provided downstream from the regulator prior to the servicing hose and mounted on the instrument panel.

3.4.7.3 Servicing chuck. A servicing chuck shall be provided that conforms to drawing 69E35397-5. If required an adapter shall be supplied to connect the servicing chuck to the air hose.

3.4.8 Blowdown receiver. A receiver shall be provided to prevent visible condensate from blowing to the atmosphere when the compressor is unloaded. The condensate receiver shall be of sufficient capacity to provide for a minimum of eight hours continuous operation at conditions of 97° F dry bulb and 87° F wet bulb.

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3.5.1.1 Intermittent duty. The compressor unit shall be capable of an intermittent duty output of 30 scfm of air at a discharge pressure of 500 psi. Operation under these conditions shall not produce adverse effects or malfunctions in the compressor unit.

3.5.2 Oil consumption. Oil consumption shall not exceed an average of one pint of oil per 25 hours of operation.

3.5.3 Low-temperature operation. At an ambient temperature of 0° F, the pressure-servicing system shall be capable of operating with a 400-psi inlet pressure to the regulator. The pressure-servicing system shall be adjusted to deliver rated capacity of the compressor at pressures of 50, 100, 200, and 300 psi for 10 minutes at each pressure.

3.5.4 High-temperature operation. At an ambient temperature of 125° F, the pressure-servicing system shall be capable of delivering rated capacity at pressures of 50, 100, 200 and 300 psi for 10 minutes at each pressure.

3.5.5 Hydrostatic pressure. The high and low pressure zones of the regulator shall be capable of operation after being subjected to a hydrostatic pressure of 750 psi.

3.6 Dimensions. The overall outside dimensions of the compressor shall not exceed 100 inches in length, 30 inches in width, and 50 inches in height.

3.7 Weight. The maximum dry weight of the compressor, including all accessories, shall not exceed 3200 pounds.

3.8 Finish. The portions of the compressor normally painted shall be cleaned, treated and painted in accordance with the manufacturers standard practice and the finish coat shall be gray, No. 16473, in accordance with FED-STD-595. The compressor shall be marked in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Classification of tests. Inspection and testing of the compressor shall be in accordance with MIL-C-26805 and those exceptions specified herein.

4.2 Preproduction test samples. At the option of the contractor, a second compressor may be used in performing the preproduction test. Should the contractor elect to use a second preproduction sample, both samples shall be representative models and both shall be subjected to the preproduction test listed below. Other test requirements not listed below may be performed on either unit, except, the initial teardown inspection, endurance test and final teardown inspection must be performed on the same sample.

Examination of product
Hydrostatic pressure test
Performance demonstration.

4.3 Preproduction testing.

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4.3.4 Pressure regulation test. The compressor pressure regulators shall be tested to insure compliance with 3.4.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. The compressor shall be preserved, packaged and packed in accordance with MIL-C-3600.

6. NOTES.

6.1 Intended use. The AF/E32A-1 compressor is intended for use in hangars, garages, paint and dope shops and any installation requiring extremely dry air or explosion proof air compressing equipment.

6.2 Ordering data.

a. Title, number and date of this specification.
b. When electrical system shall be connected other than as specified (see 3.4.10).
c. Selection of applicable levels of preservation, packaging and packing..

6.3 Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian
Air Force - 84

Preparing Activity:
Air Force - 84

Project No.
4310-F049

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-C-26165D(USAF): Compressor, Air, Electric Motor Driven, 30 CFM, 400 PSI, Skid Mounted Type AF/E 32A-1

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE. AF-WP-O-MAY 64 505

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